Alledin

A personal allergy guide

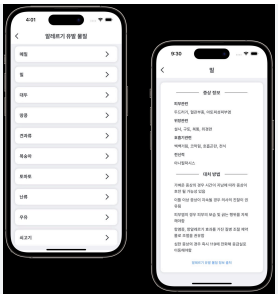
A mobile service that finds franchise restaurant menus compatible with a user's food allergies.

Period

May – Nov 2023

Role

Four-person team · one of two backend developers



What it does

- Save allergies from 21 designated allergens
- Explain symptoms, responses, and severity
- Search restaurants and menus using location and allergy data

My contribution

- Collected roughly 10,000 franchise locations, products, and allergen records, normalized them as CSV, and stored them in AWS RDS.
- Designed MVP wireframes in Figma and standardized server errors with an Exception Enum.
- Converted public-data coordinates for the map API and cleaned inconsistent records.

A problem I worked through

Problem

The service queried a user's allergies for each menu item, causing restaurant searches to take roughly one minute.

How I solved it

Changed the service to load the user's allergy data once and reuse it, reducing repeated database reads.

Process and decisions

Preparing the data

We gathered location, product, and allergen information from franchises with more than 100 branches. I removed unusable public records, normalized inconsistent names, prepared CSV files for RDS, and examined coordinate conversion for the map API.

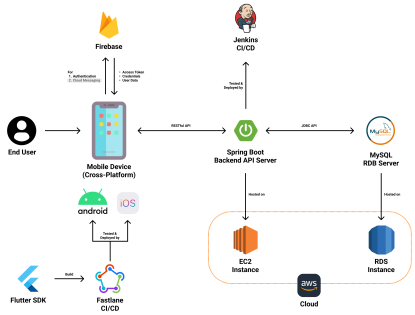
Fixing the search path

I traced the request through the controller, service, and database and found a repeated allergy query for every menu. Loading allergy data once and reusing it removed the bottleneck behind searches that took about a minute.

What remained

New and discontinued menu items change continuously. A production service would need a regular data refresh and validation workflow.

System architecture



Technology

Java 17 · Spring Boot 3 MySQL · AWS RDS AWS EC2 Figma